

Trader Performance vs. Market Sentiment

An Analysis of Hyperliquid Trading Data Against the Bitcoin Fear & Greed Index

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Executive Summary

This report examines the relationship between Bitcoin market sentiment — measured by the Fear & Greed Index — and the trading performance of 32 accounts on the Hyperliquid decentralized exchange, covering 211,218 trades between May 2023 and May 2025 across 246 traded assets.

The analysis merges two datasets: (1) daily Bitcoin market sentiment classifications, and (2) granular trade-level execution records including position direction, size, and realized profit and loss (PnL). The goal is to identify how trader behavior and profitability shift across sentiment regimes and to surface insights that can inform smarter, sentiment-aware trading strategies.

Key Findings

- **Traders were most profitable per-trade during Extreme Greed (avg. \$130.21/trade, 89.2% win rate) and Fear (avg. \$112.63/trade, 87.3% win rate) — the two emotional extremes — while Neutral sentiment produced the weakest average returns (\$71.20/trade).**
- **Win rates stayed high across all regimes (76%–89%), but profitability per dollar traded roughly tripled in Extreme Greed (2.18%) versus other regimes (0.65%–0.75%), suggesting traders extract disproportionate value when the market is euphoric.**
- **Traders flip from net long to net short as sentiment shifts from Fear to Greed: long positions made up 69% of new entries during Extreme Fear but only 45% during Extreme Greed — a textbook contrarian-turned-momentum pattern.**
- **Short positions were the more profitable side during Fear regimes (avg. \$207.88/trade in Fear) while long positions were more profitable during Extreme Greed (avg. \$61.84/trade) — performance is highly conditional on aligning position direction with the regime.**
- **Trading volume and daily PnL both had a mild negative correlation with the Fear & Greed value (-0.26 and -0.08 respectively), indicating this trader cohort was somewhat more active and profitable when sentiment cooled rather than during outright euphoria.**
- **Performance was highly concentrated: the top 5 accounts generated over \$6.3M of the \$10.25M total realized profit, while the bottom 5 accounts lost a combined ~\$254K — position sizing and risk discipline matter more than sentiment timing alone.**

1. Data & Methodology

1.1 Datasets

- Bitcoin Market Sentiment Dataset: Daily Fear & Greed Index values (0-100) and classifications (Extreme Fear, Fear, Neutral, Greed, Extreme Greed), Feb 2018 – May 2025.
- Hyperliquid Historical Trader Data: 211,224 individual trade executions from 32 accounts across 246 symbols, including execution price, size, side, direction (open/close, long/short), closed PnL, fees, and timestamps.

1.2 Processing Steps

- Parsed trade timestamps (IST) and normalized to calendar dates to join with daily sentiment classifications — 99.997% of trades matched a sentiment record.
- Separated 'closed' trades (Direction = Close Long, Close Short, Buy, Sell) which carry realized PnL from 'opening' trades (Open Long, Open Short) which do not, to avoid diluting profitability metrics with zero-PnL entries.
- Grouped the five official sentiment classes into a simplified 3-bucket view (Fear, Neutral, Greed) for high-level comparisons, while preserving the full 5-class breakdown for granular analysis.
- Computed win rate (share of closed trades with positive PnL), average and median PnL per trade, total realized PnL, trading volume, and long/short positioning bias for each sentiment regime.

1.3 Scope & Limitations

- The trader cohort is small (32 accounts) and concentrated — results reflect this specific group's behavior on Hyperliquid, not the broader market.
- Closed PnL reflects realized gains/losses only; unrealized (open) positions are not captured, so daily figures can be noisy around position rollovers.
- Correlation does not imply causation — sentiment may influence trader behavior, or both may respond to a shared underlying driver (e.g., price volatility).

2. Trader Performance by Market Sentiment

The table below summarizes trading activity and profitability across the five official Fear & Greed classifications.

Sentiment	Trades	Closed Trades	Win Rate	Avg PnL/Trade	Total PnL
Extreme Fear	21,400	10,406	76.2%	\$71.03	\$739,110
Fear	61,837	29,808	87.3%	\$112.63	\$3,357,155
Neutral	37,686	18,159	82.4%	\$71.20	\$1,292,921
Greed	50,303	25,176	76.9%	\$85.40	\$2,150,129
Extreme Greed	39,992	20,853	89.2%	\$130.21	\$2,715,171

Table 1: Trade volume and realized PnL by sentiment classification

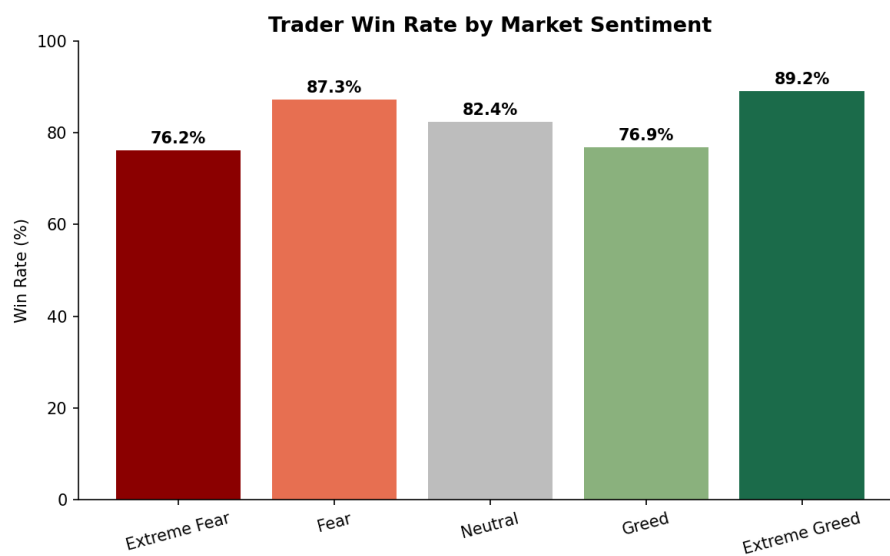


Figure 1: Win rate is highest in the two emotional extremes (Extreme Greed, Fear) and lowest in Extreme Fear

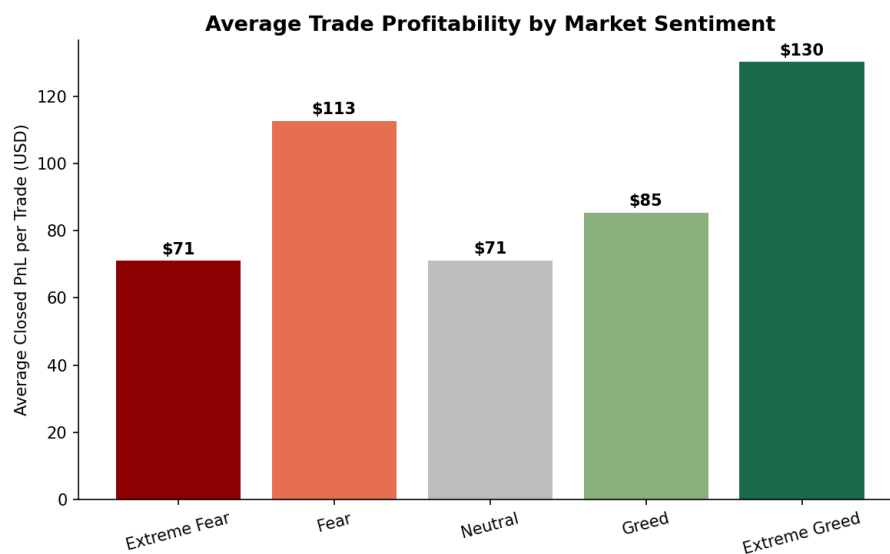


Figure 2: Average profit per trade peaks during Extreme Greed, nearly double the Neutral-regime average

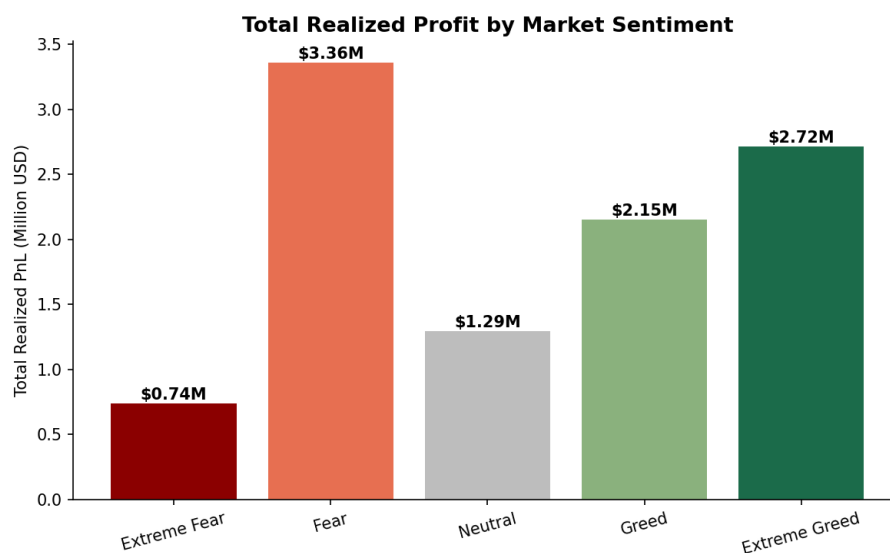


Figure 3: Fear and Extreme Greed regimes generated the largest share of total realized profit

Interestingly, profitability does not scale linearly with 'greed' the way one might expect. Fear and Extreme Greed — the two ends of the emotional spectrum — produced the strongest average returns, while Neutral sentiment consistently underperformed. This suggests the trader cohort profits most from decisive, high-conviction environments rather than ambiguous ones, and that a disciplined contrarian-to-momentum strategy (buying fear, riding greed) aligns with how this group actually trades.

3. Long/Short Positioning Behavior

Traders' directional bias shifts markedly with sentiment, revealing a clear behavioral pattern.

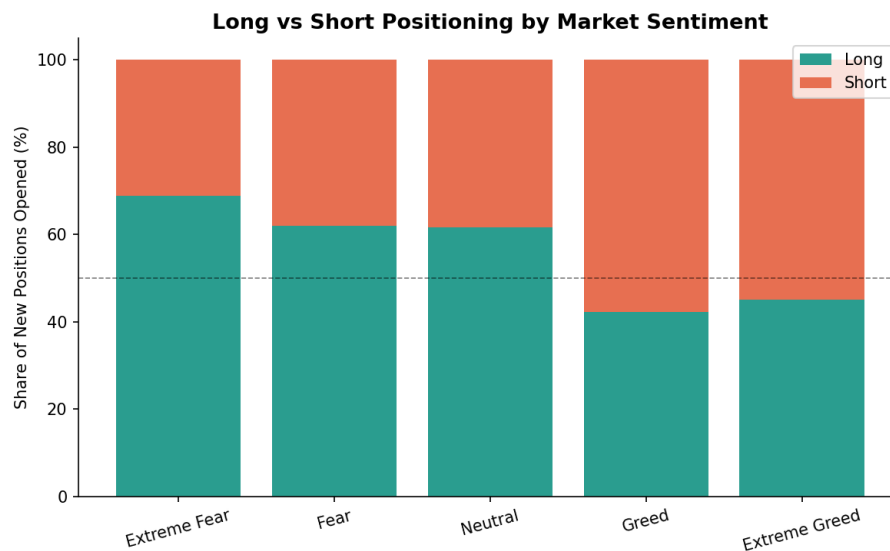


Figure 4: Share of new long vs. short positions opened, by sentiment regime

Sentiment	% Long Positions	% Short Positions
Extreme Fear	68.8%	31.2%
Fear	62.1%	37.9%
Neutral	61.7%	38.3%
Greed	42.3%	57.7%
Extreme Greed	45.1%	54.9%

Table 2: New-position long/short split by sentiment

During Extreme Fear, nearly 7 in 10 new positions are long — consistent with a 'buy the dip' contrarian approach when the market is most pessimistic. As sentiment moves into Greed, the balance flips and short positions become more common, possibly reflecting profit-taking or bets on a pullback after a run-up.

3.1 Does Direction Match Profitability?

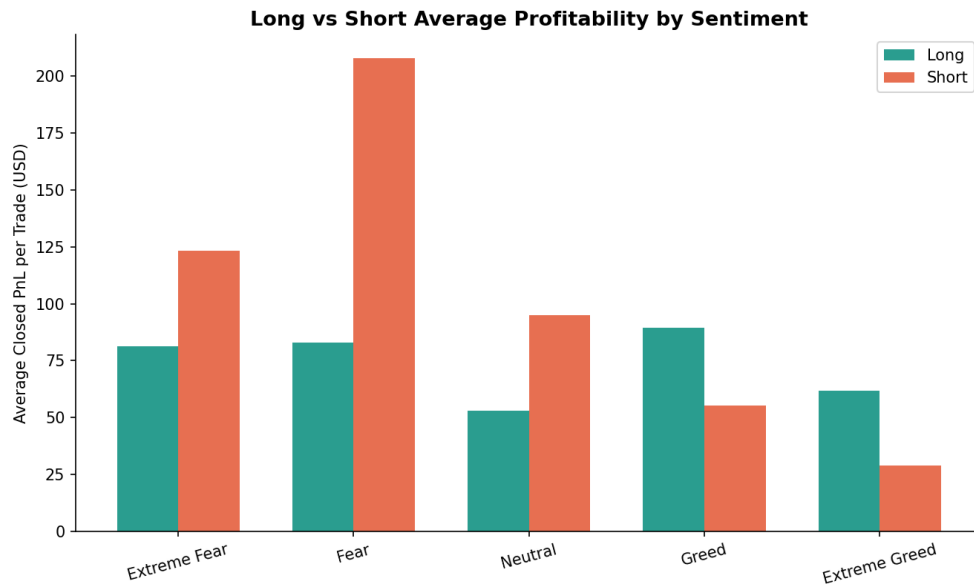


Figure 5: Average PnL per trade, split by long vs. short and sentiment regime

The most profitable alignment was going short during Fear (avg. \$207.88/trade, \$1.9M total) and going long during Extreme Fear (avg. \$81.20/trade, \$507K total) or Extreme Greed (avg. \$61.84/trade). Notably, shorting during Fear was substantially more lucrative than the prevailing long-bias behavior in that regime — traders who bucked the crowd and shorted into fear captured the largest average gains in the dataset.

4. Sentiment and PnL Over Time

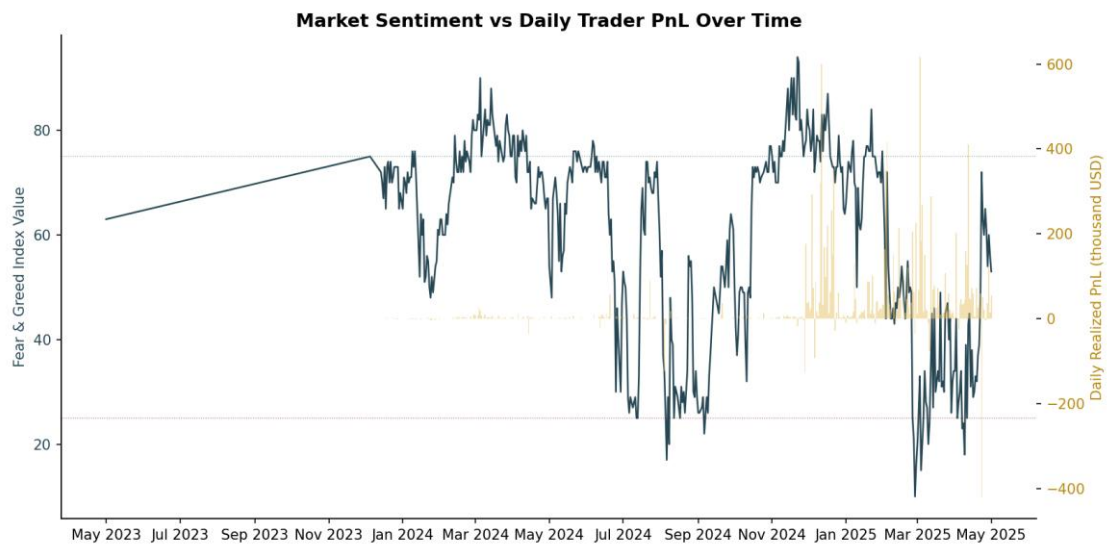


Figure 6: Daily Fear & Greed Index value (line) overlaid with daily realized PnL (bars), May 2023 – May 2025

Correlating the raw daily Fear & Greed Index value against daily trading volume and daily realized PnL shows only weak relationships: a correlation of -0.264 with volume and -0.083 with PnL. This weak-to-mild negative correlation indicates this trader cohort was, if anything, slightly more active and profitable as sentiment cooled rather than during outright euphoric spikes — reinforcing that raw sentiment score alone is a poor standalone predictor of this group's daily profitability, and that the categorical regime (which captures behavioral shifts in positioning) is more informative than the continuous index value.

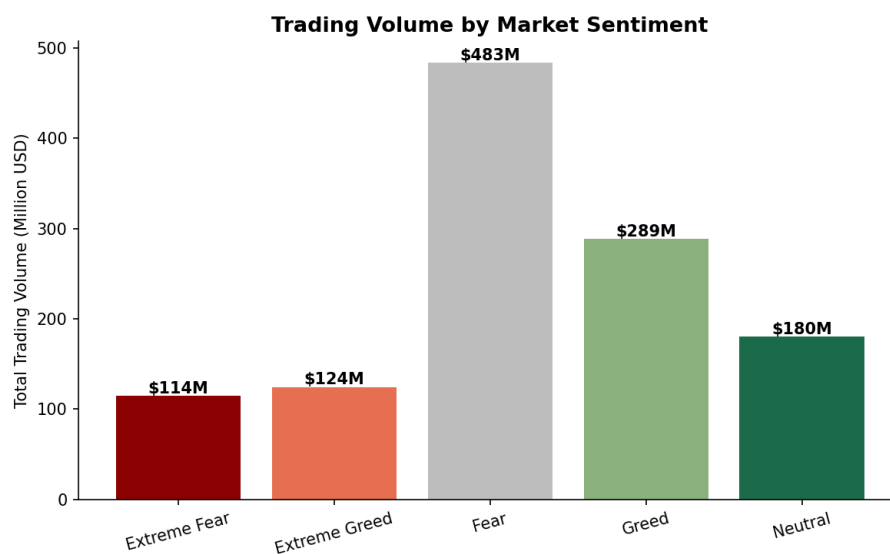


Figure 7: Total trading volume by sentiment regime — Fear periods saw the highest volume

5. Trader Concentration

Performance across the 32-account cohort is highly unequal. The table below shows the accounts with the largest realized gains and losses over the full period.

Account (truncated)	Total Realized PnL
0xb1231a4a...bfed23	\$2,143,383
0x083384f8...9012	\$1,600,230
0xbaaaf657...637864	\$940,164
0x513b8629...49c4ff1	\$840,423
0xbee1707d...437aab	\$836,081

Table 3: Top 5 accounts by total realized PnL

Account (truncated)	Total Realized PnL
0x7f4f299f...b9db8f	\$14,900
0x39cef799...5cae3c	\$14,457
0x3998f134...bbce891	-\$31,204
0x271b2809...de622ab	-\$70,436
0x8170715b...a0c0b	-\$167,621

Table 4: Bottom 5 accounts by total realized PnL

The top 5 accounts alone account for over \$6.3M — roughly 62% — of the cohort's \$10.25M total realized profit, while the bottom 5 accounts lost a combined \$254K. This concentration suggests that individual skill, position sizing, and risk management explain more variance in outcomes than market sentiment alone. Sentiment-aware strategies are likely to be a meaningful edge, but not a substitute for sound risk discipline.

6. Strategic Recommendations

6.1 Lean Into Emotional Extremes, Not the Middle

Both average and total profitability were weakest during Neutral sentiment. Strategies that scale up conviction and position sizing during Fear/Extreme Fear and Greed/Extreme Greed — and reduce activity in Neutral stretches — align with where this data shows the edge actually exists.

6.2 Consider Contrarian Positioning During Fear

Shorting during Fear regimes produced the single highest average PnL per trade (\$207.88) in the entire dataset, even though the majority of traders were positioned long. A disciplined contrarian short strategy during Fear — rather than following the long-biased crowd — merits further backtesting.

6.3 Ride Momentum During Extreme Greed, But Manage Size

Extreme Greed produced the highest win rate (89.2%) and second-highest average PnL, but also the smallest average trade size (\$3,112), suggesting traders were already de-risking into strength. This is a rational pattern worth preserving rather than overriding.

6.4 Use Regime Classification, Not Raw Index Value

Because the continuous Fear & Greed score showed only weak correlation with volume and PnL, strategies should key off the categorical regime (Extreme Fear / Fear / Neutral / Greed / Extreme Greed) and positioning shifts around those thresholds, rather than treating the index as a continuous signal.

6.5 Prioritize Risk Management Over Sentiment Timing Alone

The wide spread between top and bottom performers shows that sentiment awareness is not sufficient on its own. Position sizing discipline, stop-losses, and consistent process are likely to matter as much or more than correctly reading the sentiment regime.

Appendix: Methodology Notes

- Data sources: `fear_greed_index.csv` (Bitcoin Fear & Greed Index) and `historical_data.csv` (Hyperliquid trade executions).
- Analysis performed in Python (pandas, matplotlib) — full script provided alongside this report.
- All dollar figures reflect realized (closed) PnL in USD as recorded in the source data; fees are reported separately and not netted out of PnL figures.
- Correlation coefficients are Pearson correlations computed on daily-aggregated series.